



EIMBox - bKash Pay Bill Integration

Technical Integration Specification & API Reference Guide

Document Version: 2.0 | **Compliance:** bKash Pay Bill Specification v2.0 | **Date:** September 2026

Target Audience: bKash Technical Integration Team, MFS Aggregators, Institution Admins

1. System Overview & Architecture

The **EIMBox Educational Institution Management System** provides automated real-time student fee settlement and billing integration for **bKash Mobile Financial Services (Pay Bill Module)**.

- Guardians or students select the EIMBox Educational Institution Biller inside the **bKash Mobile App** or dial *USSD* (247#).
- The user enters their Student ID / Reference Number (`ref_id`).
- bKash queries the EIMBox **Check Bill API** (`/CheckBill`), which dynamically calculates real-time unpaid dues and returns student details and payable amount.
- Upon user confirmation and PIN verification, bKash dispatches a transaction notification to the **Bill Payment API** (`/BillPayment`), updating student finance ledgers (`stfinance`), issuing receipt vouchers (`stpr`), and logging audit trails.
- In case of network timeouts or reconciliations, bKash calls the **Recheck Bill Payment API** (`/RecheckPayment`) to verify settlement status.

INTEGRATION ENDPOINTS: Base URLs for Integration:

- Production Base URL: <https://console.eimbox.com/api/payment/bkash/v1/>
- Staging / Test URL: <http://localhost/eimbox-dashboard/eimbox-materio/api/payment/bkash/v1/>

2. Security & Authentication

All API communications must occur over secure **TLS 1.2+ (HTTPS)**.

2.1 Authentication Methods

- **HTTP Basic Authentication Header:** `Authorization: Basic <base64(username:password)>`
- **Payload / Query Credentials:** `username` and `password` passed inside JSON body or URL query parameters.

2.2 Supported Authentication Models

- **Global Aggregator Model (Default):** Central bKash Biller partner credentials (`bKash00` / `bKash12321` or `demo3` / `aaa`).
- **Multi-Tenant / Per-Institute Model:** Gateway dynamically validates credentials against the `payment_gateway_keys` table for institutions with direct merchant agreements.

3. Reference Parameter Convention

Parameter	Type	Mandatory	Description	Accepted Formats
ref_id	String (6-30)	Yes	Student Identifier (<code>stid</code>) or Combined Reference	• 10-digit Student ID: <code>1031870001</code> • Combined Delimited: <code>103187-1031870001</code>
bill_month	String (6)	Optional	Billing Month (<code>MMYYYY</code> or <code>na</code>)	• <code>092026</code> for September 2026 • <code>na</code> for cumulative dues till date
sccode	Numeric (6)	Optional	6-digit Institution Code (EIIN)	Explicit school code if not present in <code>ref_id</code>

4. API Endpoints Specification

4.1 Check Bill API (/CheckBill)

Validates student existence, calculates real-time unpaid dues from **Dues Ledger**, and returns the student profile, payable amount, and due date.

- **Endpoint:** <https://console.eimbox.com/api/payment/bkash/v1/CheckBill.php>
- **HTTP Methods:** POST

Request Parameters

Parameter	Type	Required	Description	Example
username	String	Yes	Biller Username	bKash00
password	String	Yes	Biller Password	bKash12321
ref_id	String	Yes	Student Reference ID	1031870001
bill_month	String	No	Month (MMYYYY or na)	na
amount	String	No	Optional amount parameter	1283

Sample Request (POST JSON)

```
{
  "username": "bKash00",
  "password": "bKash12321",
  "ref_id": "1031870001",
  "bill_month": "na"
}
```

Success Response (200 OK)

```
{
  "ErrorCode": "200",
  "ErrorMsg": "Success",
  "Consumer_Name": "Afruja Akter",
  "Bill_month": "na",
  "Bill_amount": "1283",
  "Bill_due_date": "2026-09-30",
  "Trxid": "CHK20260922065637288",
  "Querytime": "20260922065637",
  "Amount_Breakdown": "Class: Six (Tagor) | Roll: 1 | SC: 103187"
}
```

4.2 Bill Payment API (/BillPayment)

Settles student finance ledgers, generates a receipt voucher with an 8-digit receipt number and logs transaction audit trails.

• **Endpoint:** <https://console.eimbox.com/api/payment/bkash/v1/BillPayment.php>

• **HTTP Methods:** POST

Request Parameters

Parameter	Type	Required	Description	Example
username	String	Yes	Biller Username	bKash00
password	String	Yes	Biller Password	bKash12321
ref_id	String	Yes	Student Reference ID	1031870001
bill_month	String	No	Month (MMYYYY or na)	na
amount	Numeric	Yes	Paid Amount in BDT	1283
user_mobile_number	String	No	Payer Mobile Number	01711000000
trxid	String	Yes	bKash Unique Transaction ID	BKASH_TXN_987654
paytime	String	Yes	Timestamp (YYYYMMDD)	20260922

Sample Request (POST JSON)

```
{
  "username": "bKash00",
  "password": "bKash12321",
  "ref_id": "1031870001",
  "bill_month": "na",
  "amount": "1283",
  "user_mobile_number": "01711000000",
  "trxid": "BKASH_TXN_987654",
  "paytime": "20260922"
}
```

Success Response (200 OK)

```
{
  "ErrorCode": "200",
  "ErrorMsg": "Success",
  "Consumer_Name": "Afruja Akter",
  "Total_amount": "1283",
}
```

```

    "Trxid": "BKASH_TXN_987654",
    "Paytime": "20260922",
    "Amount_Breakdown": "Class: Six | Roll: 1 | PR: 25000101"
  }

```

4.3 Recheck Bill Payment API (/RecheckPayment)

Inquires previous settlement status by `trxid` during network drops, timeouts, or batch audit reconciliations.

- **Endpoint:** <https://console.eimbox.com/api/payment/bkash/v1/RecheckPayment.php>
- **HTTP Methods:** POST

Request Parameters

Parameter	Type	Required	Description	Example
username	String	Yes	Biller Username	bKash00
password	String	Yes	Biller Password	bKash12321
trxid	String	Yes	bKash Transaction ID to verify	BKASH_TXN_987654

Sample Success Response (200 OK)

```

{
  "ErrorCode": "200",
  "ErrorMsg": "Success",
  "PreviousMessage": "Payment Information Updated Successfully. PR: 25000101 | Amount: 1283 BDT"
}

```

5. Standard Error Codes & Handling

Error Code	Error Message	Description & Trigger Conditions	Remediation
200	Success	Transaction / inquiry completed successfully.	No action required.
201	Authentication failed	Invalid username or password in Header / Payload.	Verify API credentials.
202	Mandatory Field missing	Required field (<code>ref_id</code> , <code>username</code> , <code>password</code> , <code>trxid</code> , <code>amount</code>) is missing.	Provide all mandatory parameters.
203	Data Mismatch	Requested parameters do not match database records.	Verify student academic session.
204	Already paid	Student has 0 outstanding unpaid dues for the queried period.	No further payment required.
205	Data not found	Student record or transaction ID not found in database.	Verify <code>ref_id</code> or <code>trxid</code> .
206	Due date over	Bill payment deadline has expired.	Contact institution admin.
207	Minimum amount not paid	Paid amount is less than minimum acceptable fee.	Adjust payment amount.
208	Pay amount and biller amount not match	Paid amount does not match calculated total outstanding dues.	Query <code>/CheckBill</code> for exact dues.
209	Internal error	Database exception or unhandled server error.	Check server error logs.
210	School not found	Institution EIIN / School Code cannot be resolved.	Pass valid 6-digit EIIN.

8. Integration Testing & PowerShell Automation

```
$baseUrl = "http://localhost/eimbox-dashboard/eimbox-materio/api/payment/bkash/v1"
$headers = @{ "Content-Type" = "application/json" }

# 1. Check Bill
$chkBody = @{
    username = "bKash00"
    password = "bKash12321"
    ref_id = "1031870001"
    bill_month = "na"
} | ConvertTo-Json

$chkRes = Invoke-RestMethod -Uri "$baseUrl/CheckBill.php" -Method Post -Headers $headers
-Body $chkBody
$chkRes | ConvertTo-Json

# 2. Bill Payment
$payTxnId = "BKASH_" + (Get-Random -Minimum 100000 -Maximum 999999)
$payBody = @{
    username = "bKash00"
    password = "bKash12321"
    ref_id = "1031870001"
    bill_month = "na"
    amount = $chkRes.Bill_amount
    user_mobile_number = "01711000000"
    trxid = $payTxnId
    paytime = (Get-Date).ToString("yyyyMMdd")
} | ConvertTo-Json

$payRes = Invoke-RestMethod -Uri "$baseUrl/BillPayment.php" -Method Post -Headers
$headers -Body $payBody
$payRes | ConvertTo-Json

# 3. Recheck Payment
$recheckBody = @{
    username = "bKash00"
    password = "bKash12321"
    trxid = $payTxnId
} | ConvertTo-Json

$recheckRes = Invoke-RestMethod -Uri "$baseUrl/RecheckPayment.php" -Method Post -Headers
$headers -Body $recheckBody
$recheckRes | ConvertTo-Json
```